

ANCIENT CORAL
Compared to their soft-bodied relatives the anemones and jellyfish, corals were preserved well as fossils in rocks because of their hard skeletons, such as this 400-million-year-old fossil coral. Each coral animal formed a skeleton joining that of its neighbor to create chains with large spaces between them.



1



2



3



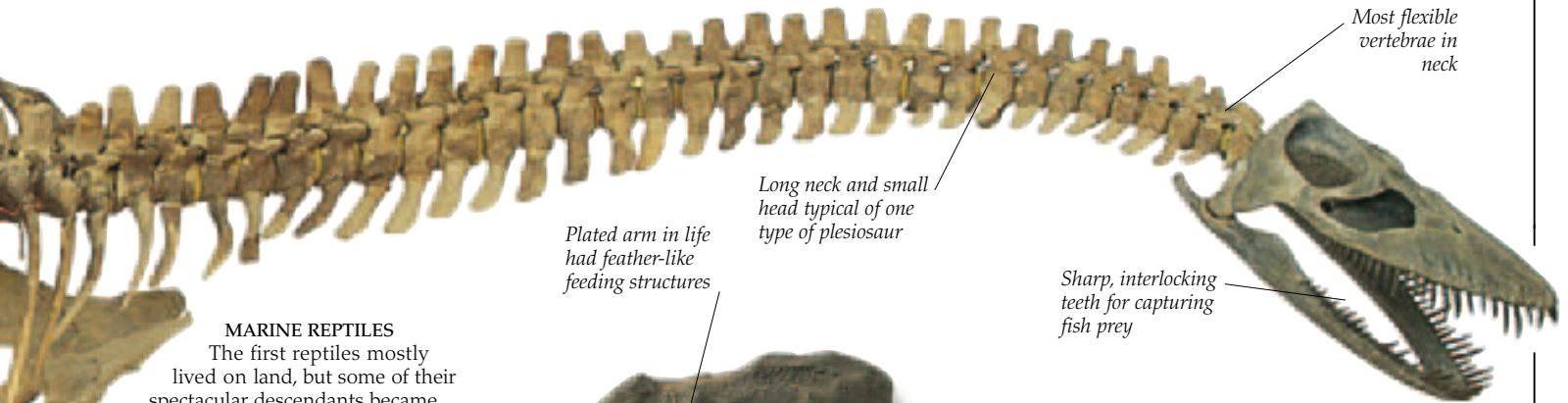
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CHANGING OCEANS

One giant ocean, Panthalassa, surrounded the supercontinent Pangaea (1), 290–240 mya (million years ago). At the end of this period, many kinds of marine life became extinct. Pangaea broke up, with part drifting north and part south, with the Tethys Sea between.

CONTINENTAL DRIFT

The northern part split to form the North Atlantic 208–146 mya (2). The South Atlantic and Indian Oceans began to form 146–65 mya (3). The continents continued to drift 1.64 mya (4). Today the oceans are still changing shape—the Atlantic Ocean gets wider by an inch or so each year.



Most flexible vertebrae in neck

Long neck and small head typical of one type of plesiosaur

Plated arm in life had feather-like feeding structures

Sharp, interlocking teeth for capturing fish prey

MARINE REPTILES

The first reptiles mostly lived on land, but some of their spectacular descendants became adapted for life in the sea. Among the best known are the plesiosaurs. They first appeared around 200 million years ago. Plesiosaurs swam using their flippers, as either oars or wings—to “fly” through the water like turtles do today. They eventually died out around 65 million years ago along with their land-based cousins, the dinosaurs. The only true ocean-dwelling reptiles today are the sea snakes and sea turtles.

Smaller, front flipper also had five elongated toes



Long, flexible stem anchored crinoid in seabed gardens

SEA LILY

A complete fossil of a sea lily (crinoid) is quite a rare find even though large numbers of these animals grew on the bottom of ancient oceans. The skeleton, composed of small bony plates, usually broke up when the animal died. Although they are far less numerous today, sea lilies are still found living below 330 ft (100 m). Sea lilies are relatives of feather stars, but unlike them are usually anchored to the seabed. Their arms surround an upward-facing mouth and are used to trap small particles of food drifting by.

All-around vision provided by large, curved eye

Segmented body allowed trilobite to roll up like a woodlouse



DEAD AND GONE

Trilobites, one of the most abundant creatures living in the ancient seas, first flourished over 510 million years ago. They had jointed limbs and an external skeleton like insects and crustaceans (such as crabs and lobsters) but they died out some 250 million years ago.